Nasief Khan

Software Engineer | TypeScript, Node.js, React, Svelte | API and HR/Payroll Integrations | AI-Assisted Development

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PROFESSIONAL SUMMARY

Software Engineer with 6+ years of software development experience, including 4+ years building and supporting production systems for a mobile-first employee engagement platform used by around 500,000 frontline workers. Delivered integrations and features for 20+ enterprise clients across retail, mining and automotive manufacturing.

Experienced across TypeScript, Node.js, Express, React, Svelte, MongoDB, PostgreSQL and SQL. Proven track record building HR and payroll integrations (Sage, PeopleSoft), leave management, workforce scheduling and production dashboards, taking each from API contract through to go-live and production support.

Known for owning production incidents, working directly with client IT teams through difficult go-lives, and delivering quickly: promoted within 8 months after cutting delivery time from a week to 2 days. Recently designed, built and launched a multi-tenant invoicing SaaS now used by 2 businesses.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, SQL, Python, HTML5, CSS3

Frontend: React, Svelte, Vite, Tailwind CSS, TanStack Query, AngularJS (legacy), EJS

Backend: Node.js, Express, REST APIs, Webhooks, Authentication, Session Management, Data Sync, Multi-Tenant Architecture

Data: PostgreSQL, MongoDB, SQL, Drizzle ORM, Row-Level Security, Firebase

Integrations: Sage, PeopleSoft, HR and Payroll Systems, Google Sign-In (OAuth), Azure Blob Storage

Quality and DevOps: Jest, Playwright, Zod, Git, GitHub Actions, CI/CD, Docker, Railway, Sentry, Bitbucket

AI-Assisted Engineering: Claude Code, Cursor, agentic workflows, spec and prompt writing, AI-assisted testing, refactoring and debugging

Also familiar with: Next.js, NestJS, Puppeteer

EXPERIENCE

Independent Software Engineer (AI-Assisted Development) | Cape Town (Remote)

Jun 2026 - Present

Building and shipping my own software products using AI-assisted development, from planning and specification through to production rollout.

- Designed, built and launched Carbon Invoices, a multi-tenant invoicing SaaS, now used by 2 businesses for day-to-day invoicing (see Projects).
- Developed an AI-assisted delivery workflow using Claude Code and Cursor: writing specifications, breaking work into small reviewable steps, and verifying every change with automated tests and manual checks.
- Retained full ownership of architecture, security, code quality and production outcomes while using AI agents for scaffolding, refactoring, testing, debugging and documentation.

Tech: TypeScript, Node.js, Express, React, PostgreSQL, Docker, GitHub Actions, Playwright, Claude Code, Cursor

Wyzetalk | Software Engineer, API Integrations | Cape Town

Jan 2023 - May 2026

Wyzetalk is a mobile-first employee engagement platform for frontline workers. I was one of 4 engineers (in a 20-developer team) responsible for building and maintaining 100+ client integrations across 20+ enterprise clients.

- Built end-to-end integrations between the Wyzetalk platform and client HR and payroll systems across a Node.js/TypeScript backend and Svelte frontend, contributing to features used by around 500,000 frontline employees.

- Delivered a company-wide leave management system that replaced messy supervisor-based processes: employees apply for leave and view their balances, and managers approve or decline requests in the app. Rolled out to around 40,000 employees across major South African retail clients.
- Delivered full payslip integrations with Sage and PeopleSoft, giving employees access to and tracking of their payslips directly in the app.
- Built work schedule and staff rotation features, including a workforce management tool for a major automotive manufacturing plant that shows daily attendance and lets managers reassign workers by skillset from the mobile app.
- Built a CSV import pipeline that loads 12-month shift schedules for mine workers, with parsing, validation and data cleansing, replacing a manual process of copying data by hand.
- Built production dashboards for mining and automotive manufacturing clients showing safety figures, output, and daily, weekly and monthly targets against quota.
- Built an employee recognition feature (kudos with notifications) and automated birthday templates, reducing client setup from days of back-and-forth to under 2 hours.
- Implemented authentication flows, webhook handlers and data-sync services on Express and MongoDB, with error handling for reliable data exchange between client systems and the platform.
- Served as the go-to engineer for production incidents, brought into live sessions with support and client teams to diagnose and resolve integration issues.
- Resolved go-live failures on Sage and retail payroll integrations where client production endpoints behaved differently from UAT, working with client IT teams to diagnose issues I could not test directly due to access and live-data restrictions.
- Validated external payloads with Zod and tested critical flows with Jest and Playwright.
- Reviewed teammates' code in a mandatory review-before-merge process and mentored a newer developer on the integrations team.
- Wrote and maintained technical documentation for client integrations, used for handover to client and support teams.
- Maintained and improved legacy AngularJS and newer Svelte applications; used Azure Blob Storage for client-specific static content.

Tech: TypeScript, Node.js, Express, MongoDB, SQL, Svelte, AngularJS, REST APIs, Webhooks, Authentication, Zod, Jest, Playwright, Azure Blob Storage, Sage, PeopleSoft, Cursor

Wyzetalk | Frontend Template Developer | Cape Town

Jan 2022 - Dec 2022

Built and maintained client-branded templates for the Wyzetalk platform before being promoted into feature development.

- Built and customised hand-coded EJS templates for 50+ client sites to their layout and functionality requirements.
- Took ownership of a large, unmaintained shared template repository and restructured it into a clear, organised system with consistent conventions, making templates easier to find, reuse and update.
- Cut delivery time for client templates from about a week to 2 days through reusable patterns and an organised workflow.
- Turned client feedback into responsive HTML, CSS and JavaScript interfaces.
- Promoted within 8 months into Svelte component and frontend feature development; first feature delivered was the automated birthday template, later rolled out across clients.

Tech: EJS, HTML5, CSS3, JavaScript, Svelte, Git

Databalk.Africa | Backend / Frontend Developer | Cape Town

Mar 2021 - Sep 2021

Worked on a housing management platform used in the Netherlands.

- Hired as a frontend developer and quickly moved into backend work on IRIS, a large new SQL-based system, working in small task-focused teams.

- Built Node.js and Express backend features and middleware connecting frontend applications to SQL-backed business logic.
- Shipped features and bug fixes in an existing production codebase.

Tech: Node.js, Express, SQL, JavaScript, REST APIs

Younglings Africa | Full-Stack Developer Intern | Cape Town

Mar 2020 - Feb 2021

Self-directed full-stack development internship.

- Built full-stack projects with React, Node.js, Express and MongoDB.
- Won a hackathon against 10 teams, building a custom Shopify storefront as part of a team.
- Adopted React hooks and functional components as they became standard, building reusable components and interactive features.
- Used Firebase for authentication and data storage.
- Worked in Git-based team workflows with code review and mentorship from senior developers.

Tech: React, Node.js, Express, MongoDB, Firebase, JavaScript, Git

CiTi | Python Learnership | Cape Town

Oct 2019 - Feb 2020

- Selected as 1 of 30 candidates for a competitive software engineering programme focused on Python.
- Built a foundation in Python programming, problem-solving and core software development principles, starting my move into professional software engineering.

Tech: Python, Git

PROJECTS

Carbon Invoices | Multi-Tenant Invoicing SaaS

2026 | Live in production on Railway | Source code available on request

An invoicing and sales platform where businesses manage multiple companies, shared clients, product catalogues, VAT and PDF invoices. Used by 2 businesses since November 2026.

- Designed a multi-tenant architecture using PostgreSQL row-level security: each request runs under a restricted database role, so one business can never access another's data.
- Built secure authentication: server-side sessions, scrypt password hashing, Google sign-in, rate-limited login, email invites and password resets with hashed one-time tokens, and role-based access.
- Ensured financial correctness with exact-decimal money handling, database constraints that reject invalid totals or VAT, and invoice snapshots so historical records never change.
- Generated PDF invoices and statements with Puppeteer, with three invoice designs and image processing for company logos.
- Set up testing and delivery: 34 server test suites on an isolated test database, Playwright end-to-end tests of daily workflows, GitHub Actions CI on every pull request, automated database backups, Docker builds and Sentry error monitoring.
- Built using AI-assisted development (Claude Code, Cursor), with every change specified, reviewed and tested by me.

Tech: TypeScript, Node.js, Express, React, Vite, Tailwind CSS, TanStack Query, PostgreSQL, Drizzle ORM, Zod, Puppeteer, Docker, GitHub Actions, Playwright, Sentry, Railway

EDUCATION AND TRAINING

- **National Certificate: IT Systems Development (NQF 5)**, CTU Training Solutions
- **Full Stack Web Developer Bootcamp (MERN)**, HyperionDev

INTERESTS

Speedcubing (Rubik's cubes), welding, gaming and cars.